

PAPER237: UQFFSource10 Catalogue — Master Buoyancy Integral $F_{U\backslash Bi\backslash i}$ and 26-Layer Triadic g_{UQFF}

Author: Daniel T. Murphy

Session: 59 (grokshare8d951e12.txt second-pass — Source10 Text Module)

PAPER237: UQFFSource10 Catalogue — Master Buoyancy Integral $F_{U\backslash Bi\backslash i}$ and 26-Layer Triadic g_{UQFF}

Author: Daniel T. Murphy **Framework:** UQFF v4.9 (Star-Magic) **Session:** 59 (grokshare8d951e12.txt second-pass — Source10 Text Module) **Date:** March 2026 **Classification:** Novel UQFF Catalogue — Master Buoyancy Force + 26-Layer Vectorised Triadic Gravity **Status:** Proof-Quality Whitepaper **CP3 Class:** UQFFSource10CatalogueCalculator

Abstract

This paper documents the UQFFSource10 central catalogue module, encoding the complete master buoyancy integral $F_{U\backslash Bi\backslash i}$ and 26-layer Triadic UQFF gravity $g_{UQFF}(r, t)$. The Source10 module serves as the primary reference implementation for all five UQFF force classes: low-energy nuclear reaction (LENR), dark energy expansion, magnetic resonance, relativistic buoyancy, and the 26-layer gravitational hierarchy. Validation example: Eta Carinae yields $F_{U\backslash Bi\backslash i} \approx 2.11 \times 10^{208}$ N. The module also introduces a configurable scaling_factors map and mt19937 RNG batch compute architecture for multi-system catalogue runs.

1. Physical System

Source10 is a general-purpose **astrophysical catalogue module** (not tied to a single object). The validation example uses Eta Carinae parameters:

Parameter	Value
Mass M	$\sim 150 M_{\odot} = 2.984 \times 10^{31}$ kg
Radius r	1.0×10^{14} m
$F_{U\backslash Bi\backslash i}$	$\approx 2.11 \times 10^{208}$ N
g_H (UQFF hydrogen g-factor)	1.252×10^{46}

2. Master Buoyancy Force $F_{U\backslash Bi\backslash i}$

$$F_{U\backslash Bi\backslash i} = I_{\rm grav} \cdot x_2 + F_{\rm LENR} + F_{\rm DE} + F_{\rm res} + F_{\rm rel}$$

where:

$$I_{\rm grav} = \frac{G M}{r^2}, \quad x_2 = \text{buoyancy balance parameter}$$

2.1 LENR Component

$$F_{\rm LENR} = s_{\rm LENR} \cdot \left(\rho_f v^2 \right) \cdot \left[1 + \frac{E_{\rm act}}{k_B T} \right] \cdot e^{-t/\tau_{\rm LENR}}$$

Low-energy nuclear reaction term: kinetic pressure scaled by nuclear activation threshold and temporal decay.

2.2 Dark Energy Expansion Component

$$F_{\rm DE} = s_{\rm DE} \cdot \frac{\Lambda c^2}{3} \cdot r$$

Cosmological constant Λ produces a radially-growing force term (de Sitter expansion analogy).

2.3 Magnetic Resonance Component

$$F_{\rm res} = s_{\rm res} \cdot \frac{B^2 V}{2\mu_0} \cdot \frac{\rho_n}{\rho_{\rm ref}}$$

Magnetic energy density modulated by neutron matter fraction $\rho_n/\rho_{\rm ref}$.

2.4 Relativistic Buoyancy Component

$$F_{\rm rel} = s_{\rm rel} \cdot \frac{Mc^2}{r} \cdot (1 + f_{\rm TRZ})$$

Rest-mass energy divided by radius, enhanced by time-reversal zone factor $f_{\rm TRZ}$.

3. 26-Layer Triadic UQFF Gravity

$$\boxed{g_{\rm UQFF}}(r,t) = \sum_{i=1}^{26} \left(U_{g1,i} + U_{g2,i} + U_{g3,i} + U_{g4,i} \right) + \frac{\Lambda c^2}{3} + \frac{\hbar}{\sqrt{\Delta x \Delta p}} \cdot \int \psi \, d^3x \cdot \frac{2\pi}{t_H}$$

Each layer i contributes four Triadic U_g terms:

Term	Formula	Physics
$U_{g1,i}$	$G M_i / r^2$	Newtonian gravity per layer
$U_{g2,i}$	$Q^2 / (4\pi \epsilon_0 M_i r^2)$	Charge-gravity coupling
$U_{g3,i}$	$\omega_i^2 r$	String rotation acceleration
$U_{g4,i}$	$f_{\rm vac} c^2$	Vacuum concentration

where $M_i = M/26$ (uniform layer mass distribution).

Cosmological correction: $\Lambda c^2/3$ (same as $F_{\rm DE}$ per unit length)

Quantum correction: $\hbar/\sqrt{\Delta x \Delta p} \cdot \int \psi \, d^3x \cdot 2\pi/t_H$
(Heisenberg-uncertainty-derived quantum gravity term scaled by Hubble time)

4. Configurable Architecture (Upgraded Source10)

The upgraded `UQFFSource10.h` (OpenMP version, lines 6400+) introduces:

scaling_factors **map** — per-system overrides for $s_{\rm LENR}$, $s_{\rm DE}$, $s_{\rm res}$, $s_{\rm rel}$

mt19937 **RNG** — stochastic parameter perturbation for ensemble simulations

OpenMP parallelization — 1000+ system batch compute with profiling

loadConfig(filename) — file-based configuration for reproducible runs

5. Novel Contributions

Unified 5-component master buoyancy — $\text{LENR} + \text{DE} + \text{resonance} + \text{relativistic} + \text{base}$ all in one formula

26-layer vectorised Triadic gravity — complete generalisation of single-layer Ug models

Quantum Heisenberg correction to gravity — $\hbar/\sqrt{\Delta x}, \Delta p \cdot \int \psi$ unique term

$g_H = 1.252 \times 10^{46}$ — UQFF hydrogen g-factor (47 orders above standard g_p ; used in DPM resonance)

Eta Carinae benchmark — $F_{U_Bi_i} \approx 2.11 \times 10^{208}$ N (extreme stellar-mass system validation)

6. CP3 Implementation

```
calc = UQFFSource10CatalogueCalculator()
result = calc.compute({
'M': 2.984e31, 'r': 1e14, 't': 0,
'B': 1e-3, 'volume': 1e30, 'T_temp': 1e7,
})
# result['F_U_Bi_i'] – master buoyancy
# result['g_UQFF'] – 26-layer Triadic gravity
# result['g_layers_26'] – sum of all layer contributions
```

References

Murphy, D.T. (2025). *Source10 UQFF Catalogue Module*, uqffSource10 text module + uqffSource10.h (OpenMP upgraded)

grokshare8d951e12.txt, Source10 Text Module, lines 5903–6662

Eta Carinae: Humphreys & Davidson (1994), ApJ 232, L45; Davidson & Humphreys (1997) ARA&A 35, 1

UQFF 26-layer framework: PAPER_096 (TwentySixLevelPolynomialHierarchyFullCalculator), SOURCE115
